AKSHAY VYAS

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First author at **AAAI 2026** and **NeurIPS 2026 (Spotlight, top 1% of 30,709 submissions)**. PhD candidate (coursework complete Dec 2026; available full-time from late Dec 2026) working on **how we measure and trust learned systems**: human-anchored evaluation of metrics, LLM/VLM judges, and reward models under optimization pressure, and valid-by-construction guarantees for inference. Nearly 3 years of building production software at Swiggy, Microsoft, and Goldman Sachs.

EDUCATION

The University of Texas at Dallas, Richardson, TX — Ph.D., Computer Science · GPA 3.95/4.0 *Aug 2021 – Present*
M.S., Computer Science (earned en route to Ph.D.) · GPA 4.0/4.0

IIIT Hyderabad, India — B.Tech., Electronics & Communication Engineering · GPA 3.55/4.0 *Aug 2015 – May 2019*

RESEARCH EXPERIENCE

Graduate Research Assistant · UT Dallas · Advisor: Professor Nicholas Ruozzi · NSF Grant #2327245 *Jan 2023 – Present*

Invariants as diagnostic probes for evaluation metrics | *AAAI 2026 (17.6% acceptance)*

- Formalized approximate task invariants and used them to test metric reliability without human evaluators.
- Three tasks trained from scratch: image colorization, German-English machine translation, image captioning.
- Seven baselines retrained with and without invariants: ChromaGAN, ColorFormer, DDColor, BiBERT, Bi-SimCut, SmallCap, ClipCap.
- Auxiliary losses from frozen SAN, MiDaS, BERT-NER, DistilBERT, and scene-graph nets; no new labels.
- FDD improves 3-33%, FID 7-16%, FBD 13-15%, FBD-4 5-13%; PSNR, BLEU-4, METEOR, CIDEr move under 2%.
- Showed DINOv2 features are far more invariant-sensitive than Inception, arguing for richer encoders.

Human vs. automatic-judge sensitivity: humans, LLM/VLM judges, and reward models | *Manuscript in preparation*

- Measures which semantic properties humans insist on preserving and which automatic judges silently permit.
- Graded corruption ladders over entities, relations, negation, numbers, and sentiment as causal probes.
- Identical frozen trials served to humans, metrics, LLM/VLM judges and reward models, decoded deterministically.
- Judges span BLEU, chrF++, COMET/xCOMET, CIDEr, SPICE, CLIPScore, Qwen2.5-VL, Llama-3.2-Vision, ArmoRM.
- Under MBR and best-of-N pressure, pilot results separate reward models and VLM judges from CLIP.

Certified neural upper bounds for discrete inference | *NeurIPS 2026 Spotlight (top 1%); follow-up under review*

- First learned MAP upper-bound predictor whose validity is enforced by architecture, not post-hoc calibration.
- Constrains network outputs to a provably feasible region of the LP dual, by construction.
- Holds for trained, untrained, and adversarial outputs; smooth dual objective, no LP solver or optimal labels needed.
- Warm starts tighten the JGLP dual gap by up to six orders of magnitude within 100 iterations, across 15 benchmark models.
- Test-time adaptation lands within 1e-7 of the exact optimum on 99% of constrained queries.
- Follow-up (under review): monotone, calibrated upper bounds inside AND/OR branch-and-bound, 5-10x tighter than mini-bucket.

Procedural error understanding and AR task guidance | *NeurIPS 2024 D&B; DMLR @ ICML 2023*

- Co-built CaptainCook4D, a large-scale egocentric dataset of procedural activities with annotated errors.
- Benchmarked vision and language models for procedural error detection; 50+ citations to date.

Multimodal perception for AR task guidance (DARPA PTG, demoed at MIT 2023) | *IEEE VR 2024 Workshop*

- Built a Detectron2 kitchen-object detection pipeline with training, inference, and Firebase retrieval.
- Open-set detection with Grounding DINO, plus CLIP and PROTO-CLIP for class refinement.
- Real-time segmentation with SAM and MobileSAM; 3D spatial mapping for object localization in AR.
- Co-localization fusing HoloLens 2 depth with SIFT matching in a real-time ROS pipeline.

PUBLICATIONS

† first author; longer author lists truncated at A. Vyas.

A Novel Approach to Evaluating Evaluation Metrics for Multi-Output Structured Prediction **AAAI 2026**
† **A. Vyas**, A. Pimienta, N. Ruozzi · 17.6% acceptance · Code: github.com/akshay25vyas/invariants-metric-eval

Neural Dual Bounds: Valid-by-Construction JGLP Warm-Starts for MAP and Constrained MAP **NeurIPS 2026 Spotlight**
† **A. Vyas**, S. Arya, B. Malhotra, T. Rahman, V. Gogate, N. Ruozzi

Learning to Bound: Monotonic, Calibrated Neural Upper Bounds for MPE Inference **Under review**
† **A. Vyas**, H. Dong, S. Arya, B. Malhotra, T. Rahman, V. Gogate, N. Ruozzi

CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities **NeurIPS 2024**
R. Peddi, S. Arya, B. Challa, L. Pallapothula, **A. Vyas**, et al. · Datasets & Benchmarks track

Put on Your Detective Hat: What's Wrong in This Video? **DMLR @ ICML 2023**
R. Peddi, S. Arya, B. Challa, L. Pallapothula, **A. Vyas**, et al.

Predictive Task Guidance with AI in Augmented Reality **IEEE VR 2024 Workshop**
B. Rheault, S. Arya, **A. Vyas**, et al.

Camera Model Identification Based on Local Co-Occurrence Features **IEEE SP Cup 2018**
P. Parkhiya, N. Reddy, **A. Vyas**, et al.

INDUSTRY EXPERIENCE

Software Development Engineer · Swiggy (Bundl Technologies) · Bengaluru, India *Jun 2019 – Aug 2021*

- Owned high-level and low-level architecture of ingress service relaying customer orders into cloud kitchens.
- The service carried the live order path at one of India's largest food delivery platforms.
- Ran service-level testing for the Spring Boot service by containerizing external services and databases.
- Shipped a fraud-detection REST API for delivery executives using OTP confirmation and geo-signals.
- Stack: Java (Spring Boot), Golang, Kafka, Docker, AWS (Lambda, Step Functions, DynamoDB), MySQL.

Software Development Engineer Intern · Goldman Sachs · Dallas, TX *Jun 2022 – Aug 2022*

- Built an OCR pipeline converting PDF financial tables into structured data for a banker-facing portal.
- Shipped the dashboard front end with AG Grid for interactive review (React, JavaScript, Python).

Software Development Engineer Intern · Microsoft (Bing Places) · Hyderabad, India *May 2018 – Jul 2018*

- Trained an automated spam-business classifier over local and global Bing data pipelines.
- Engineered and de-biased features; benchmarked classifier precision and recall. (C#, Model Tracker, MySQL)

TECHNICAL SKILLS

Machine learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, NumPy/SciPy, pandas, Matplotlib, OpenCV, CVXPY

Evaluation & alignment: LLM and VLM judges (Qwen2.5-VL, Llama-3.2-Vision, Gemini); reward models (ArmoRM, Skywork-Reward); best-of-N and MBR selection; reward overoptimization analysis; forced-choice preference elicitation and inter-annotator agreement.

Metrics: COMET / xCOMET, BLEU, chrF++, METEOR, BERTScore, CIDEr, SPICE, CLIPScore/RefCLIPScore, FID, FDD (DINOv2), CMMD, LPIPS, PSNR, Fréchet BERT Distance.

Vision & multimodal: Detectron2, SAM / MobileSAM, Grounding DINO, CLIP, DINOv2, ROS, OpenCV, HoloLens 2, Firebase.

Methods: Multi-output structured prediction, multi-task and auxiliary-loss training, probabilistic graphical models, LP and Lagrangian duality, branch-and-bound and AND/OR search, amortized and variational inference.

Languages: Python, C++, Java, Golang, SQL, JavaScript, Bash.

Systems: Docker, Kafka, Redis, Spring Boot, Flask, SQLite, React, AWS (Lambda, Step Functions, EC2, S3, DynamoDB, RDS), Git, Linux, Prometheus/Grafana.

RESEARCH ENGINEERING

Human studies and evaluation

- Web platform for human raters: Flask/SQLite, drag-and-drop ranking, autosave, built-in attention checks.
- Mixed-effects models, clustered bootstrap, and permutation tests over trial-level data.

Model training

- Implemented five invariant losses against frozen nets: Dice on masks, SILog on depth, KL on NER and sentiment logits.
- Swept only the invariant weight, holding every baseline hyperparameter fixed so metric movement was attributable.
- Full training matrix on single A100/V100 GPUs across seven datasets and three modalities.

Solver integration

- Integrated a neural bound predictor into a C++ branch-and-bound solver, queried at every search node.
- Optimized the Python-to-C++ inference path to millisecond latency across millions of nodes per search.
- End-to-end PyTorch training with CVXPY/Clarabel exact-LP references across UAI benchmark families.

AWARDS, SERVICE & TEACHING

- MVP award, Kitchen Intelligence & Optimization team, Swiggy (2021); under 3% of engineers per cycle.
- **Reviewer:** NeurIPS 2026.
- **Teaching Assistant, UT Dallas:** CS 6347 Statistical Methods in AI & ML; CS 4375 Introduction to Machine Learning; CS 6364 Artificial Intelligence; CS 4365 Introduction to Artificial Intelligence.
- **Teaching Assistant, IIIT Hyderabad:** Electronic Workshop II.
- Gold Medal, Men's Singles Table Tennis, National Inter-College Sports Fest (India); Sports Captain, IIIT Hyderabad.